Graduate Analysis 1: midsem exam

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Definition 1. *A sequence of functions f_n converges almost uniformly to f if for every $\epsilon > 0$ there is a set A_ϵ with $\mu(A_\epsilon) < \epsilon$ such that f_n converges uniformly to f on $(A_\epsilon)^c$*

Definition 2. *f is integrable if $\int |f| < \infty$*

Definition 3. *A sequence of integrable functions f_n is **uniformly integrable** if for all $\epsilon > 0$ there is $\delta >$ such that $\int_A |f_n| < \epsilon$ for all sets A with $\mu(A) < \delta$*

Claim 1. *On a finite measure space (X, Σ, μ), convergence almost everywhere is equivalent to convergence almost uniformly*

Proof. Consider the following:

- See Egorov's theorem in Axler for a proof of the forward implication
- Let $B \in \Sigma$ denote the points where convergence does not occur. Then $B \subseteq A_\epsilon$ which implies that $\mu(B) = 0$

□

Claim 2. *Suppose $f_n^0, f_n^1, f_n^2, f_0, f_1, f_2$ are functions such that*

- *f_n^i, f_0, f_2 are integrable*
- *$f_n^0 \leq f_n^1 \leq f_n^2$*
- *$f_n^i \to f_i$*
- *$\int f_n^0 \to \int f_0$ and $\int f_n^2 \to \int f_2$*

Then we have

- *f_1 is integrable*
- *$\int f_n^1 \to \int f_1$*

Proof. Consider the following:

- Taking limits we have $f_0 \leq f_1 \leq f_2$. Therefore $|f_1| \leq |f_0| + |f_2|$ which shows that f_1 is integrable.
- Define the following:
  - $g_n = f_n^1 - f_n^0$
  - $h_n = f_n^2 - f_n^1$
  - $g = f_1 - f_0$
  - $h = f_2 - f_1$
- By Fatou's lemma $\int g \leq \liminf \int g_n$ and $\int h \leq \liminf \int h_n$
- We have

$$\begin{aligned} \limsup \int g_n &= \limsup \int (g_n + h_n - h_n) \\ &= \int f_2 - \int f_0 - \liminf \int h_n \leq \int g \end{aligned}$$

- Hence $\int g_n \to \int g$
- We have $\int f_n^1 = \int g_n + \int f_n^0 \to \int g + \int f_0 = \int f_1$

□

Claim 3. *Suppose $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable and satisfies:*

- *$x \to f(x_0, x)$ is integrable for all x_0*
- *$g(x, y) = \partial_x f(x, y)$ exists everywhere*
- *There is an integrable real valued function h on $\mathbb{R}$ such that $|g(x, y)| \leq h(y)$*

Let $\phi(x) = \int f(x, y)dy$. Then $\phi'(x) = \int g(x, y)dy$

Proof. Consider the following:

- Fix $x \in \mathbb{R}$ and define $q(t, y) = \frac{f(x+t,y)-f(x,y)}{t}$ for $t \neq 0$
- By the mean value theorem $q(t, y) = g(\alpha(t, y), y)$
- Pick $t_n \neq 0$ such that $t_n \to 0$
- We have $\frac{\phi(x+t_n)-\phi(x)}{t_n} = \int q(t_n, y)dy$
- We have $q(t_n, y) \to g(x, y)$ for every y
- By DCT $\phi'(x) = \int g(x, y)dy$

□

Claim 4. *Suppose X is a finite measure space. If $f_n : X \to [-\infty, \infty]$ are uniformly integrable and converge pointwise to $f : X \to [-\infty, \infty]$, then we have:*

- *f is integrable*
- *$\int f_n \to \int f$*

Claim 5. *Suppose X is a random variable with finite expectation and $F \subseteq \mathbb{R}$ is closed. Suppose that whenever $P(E) > 0$ we have $\frac{\mathbb{E}[X 1_E]}{P(E)} \in F$. Then $P(X \in F) = 1$*

Claim 6. *Define the Cantor set as in 2.74 of Axler. Define $f = \sum_{n=1}^{\infty} n 1_{G_n}$. Then we have*

- *f is measurable*
- *$\int f = 1$*

Claim 7. *Suppose $E \subseteq [0,1] \times [0,1]$. If $\lambda\{x : \lambda(E_x) \leq \frac{1}{2}\} = 1$ then $\lambda\{y : \lambda(E^y) = 1\} \leq \frac{1}{2}$*